

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

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Where cancers are caught, and whether anyone looks

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Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

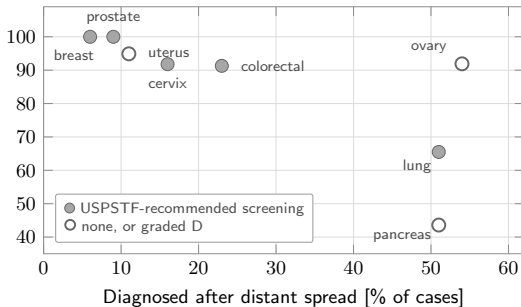
Against Moshkin and the field

From currents to a score

References

Backup

5-year survival if caught localized [%]



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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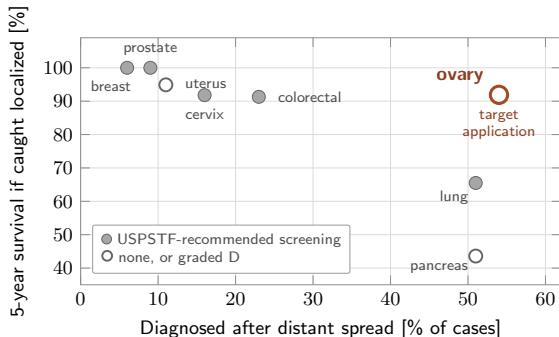
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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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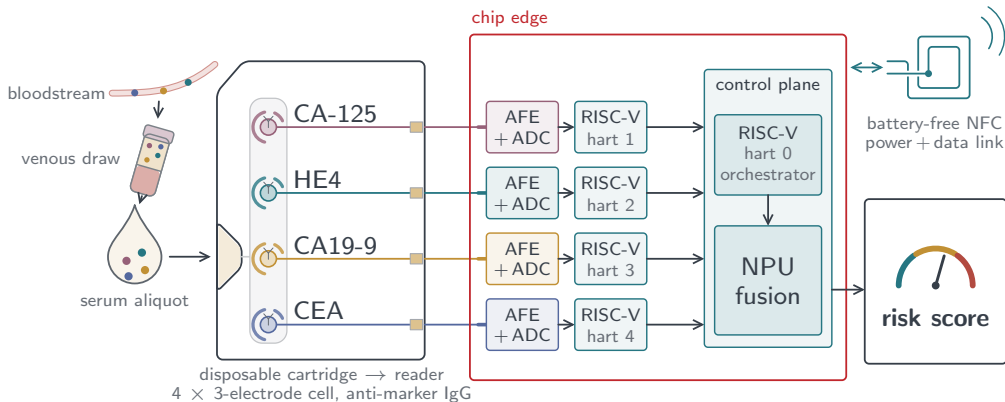
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

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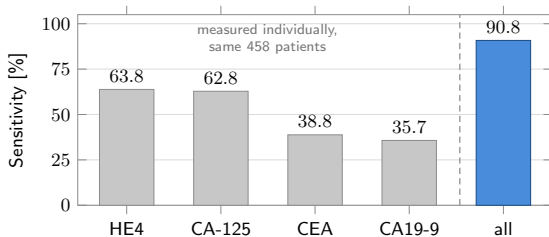
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Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Block diagram

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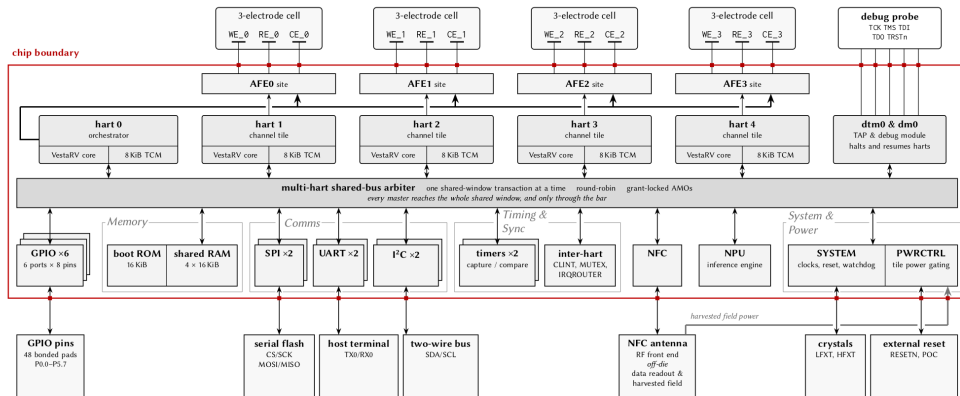
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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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ARTWORK PENDING



**ARTWORK
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Grounded-WE two-amplifier potentiostat

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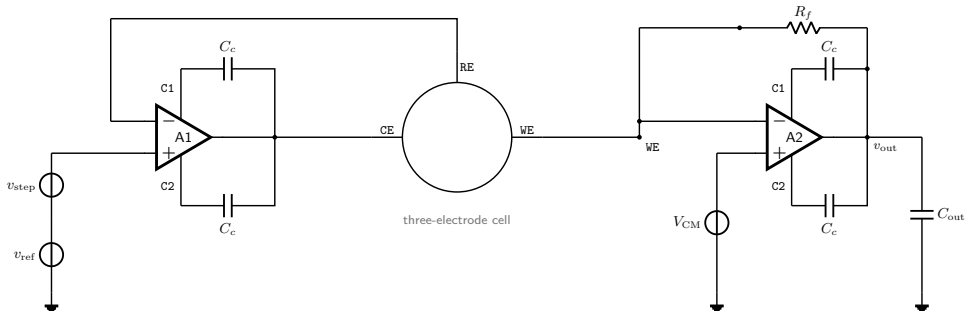
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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

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Virtuoso export of the analog top level: four channels, shared bias, test bus

The electrode model, in Verilog-A

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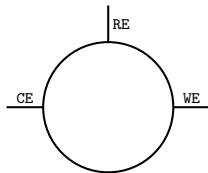
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R_u , R_{ce} , C_{dl} lumped; the faradaic branch Z_f is the Verilog-A element.

i_f faradaic current
 η overpotential, $E - E^{0'}$
 f F/RT
 $C_{O,R}(0, t)$ surface conc. of O, R
 C^* bulk conc.
 δ Nernst layer

Kinetics $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$
 Transport $\partial C / \partial t = D \partial^2 C / \partial x^2$, finite-volume, 28 expanding slabs per species
 Boundaries flux = $\pm i_f / nFA$ at the surface; $C(\delta, t) = C^*$ at the Nernst layer

Parameter	Symbol	Default	Unit	Quantity
E0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_O^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_R^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

Emulating the four markers

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Marker	Reporter	$E^{0'}$ (V)	Peak (nA)	C_R^* (μ M)	Cut-off
CA-125	methylene blue	-0.15	856	279	35 U/mL [5, 8]
HE4	ferrocene	+0.10	1311	394	70 pmol/L [6]
CA19-9	TMB / HRP [20]	+0.25	1659	458	37 U/mL [5]
CEA	anthraquinone	+0.40	939	303	5 ng/mL [5, 14]

$E^{0'}$ reporter formal potential, i.e. peak position; chosen in the -0.6 to $+0.4$ V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5 μ A on screen-printed electrodes [14, 15, 21]; C_R^* fitted to reach it

k^0 0.1 cm/s, reversible; nothing published on carbon for these markers

Shared $\alpha = 0.5$, $n = 1$, $D = 5 \times 10^{-6}$ cm²/s, $A = 0.01$ cm², 37 °C, $R_f = 560$ k Ω

Representative values, not marker-measured: these slides demonstrate the channel, not an assay.

Channel range, assay range, and where the panel was placed

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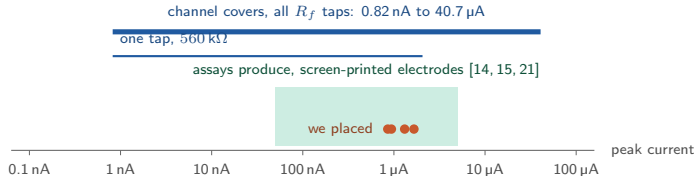
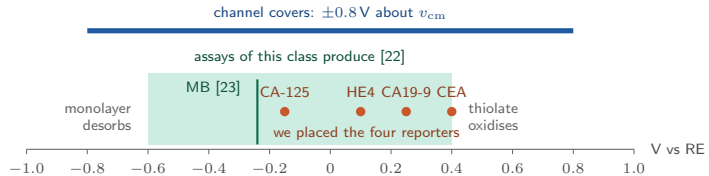
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Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then C_R^* fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

Exercising the cell: differential-pulse voltammetry

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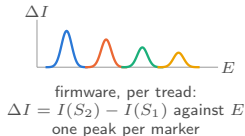
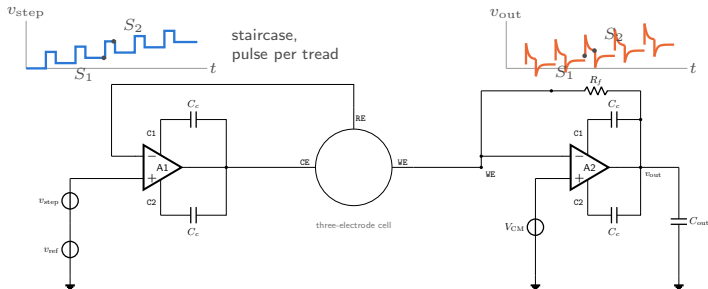
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Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$v_{out} = v_{cm} - i_{WE} R_f$

Sample

SAR at S_1 (before the pulse) and S_2 (end of the pulse), once charging has decayed

Resolve

firmware: $\Delta I = I(S_2) - I(S_1)$ vs E ; peak position: marker, peak height: concentration

Four markers resolved in one differential-pulse sweep

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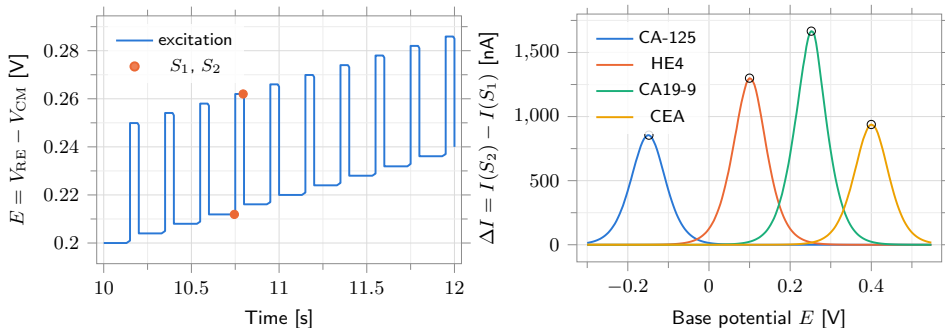
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Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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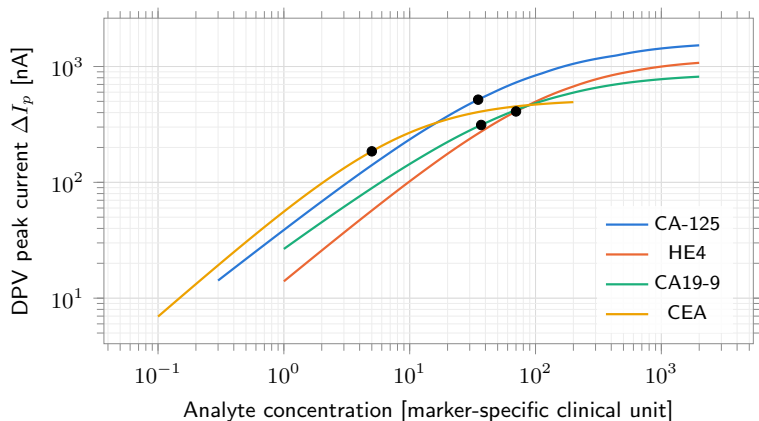
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CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Cyclic voltammetry through the assembled channel

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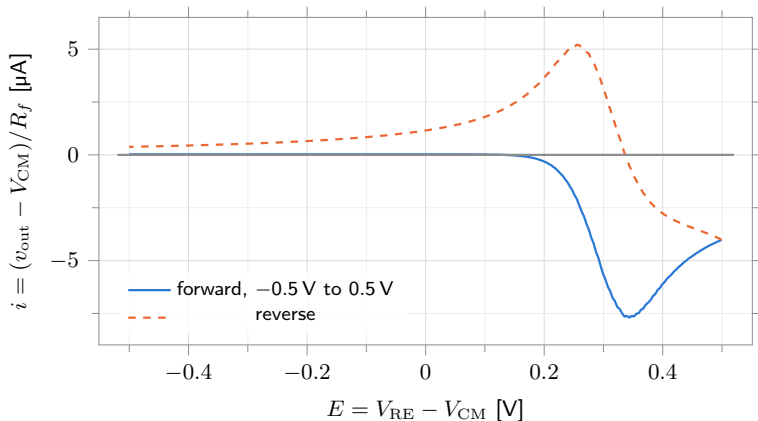
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Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Why impedance: for electrodes that stay in

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cartridge



single-shot sandwich immunoassay, incubate and read once

wearable

continuous glucose monitors, 10–15 days [27]
wound patch [28]; aptamer sensor, one week in vivo [29]



implant

Eversense implanted glucose monitor, 180–365 days [27]



1 min

1 h

1 day

1 week

1 month

1 year

time the electrode spends in the sample

EIS gives

each site's $R_{ct} \parallel C_{dl}$ arc, trended over the electrode's life, from the same channel
Firmware re-ranges, recalibrates, or retires the site

Drift is systematic: averaging removes noise, not bias [25].

How impedance is measured: the same loop, no extra hardware

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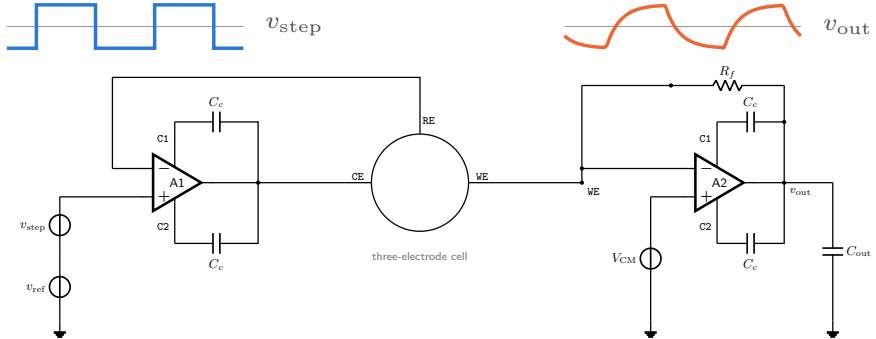
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Excite
Sense
Digitise
Solve

DAC: a small two-code square about the DC setpoint v_{ref}
 v_P across the cell; $i_{WE} = -v_{out}/R_f$
the SAR records both; firmware I/Q demodulation
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$, or a ratio to Z_{cal}

Magnitude and phase from codes: I/Q demodulation

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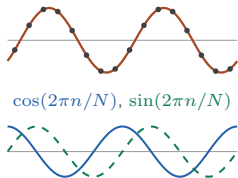
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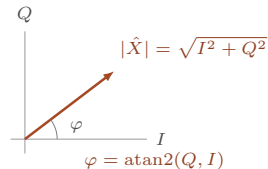
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record c_n , N per period, M periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



Records v_P and v_{out}
Divide $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$
Ratio against Z_{cal} : R_f drops out

The four loads behind the EIS plots

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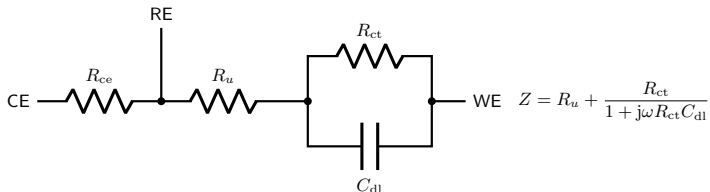
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Load	R_{ct}	C_{dl}	Gain code	Stands for
resistor	–	–	$N = 0$	an 11 k Ω resistor in place of the cell: no interface, so the trace is the channel's own error
fouled	10 k Ω	10 μ F	$N = 0$	a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface
typical	100 k Ω	1 μ F	$N = 20$	an electrode in use
fresh	1 M Ω	100 nF	$N = 63$	a new, clean surface

$R_u = R_{ce} = 1 \text{ k}\Omega$ throughout. Fouling: R_{ct} falls and C_{dl} rises through an electrode's life; tracking that drift is what EIS is for here.

Impedance spectroscopy on the same channel

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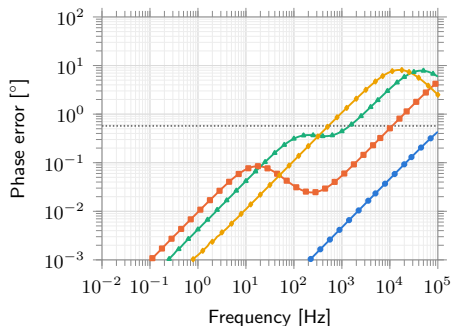
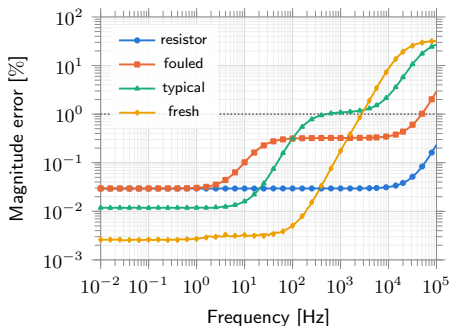
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Accuracy $\pm 1\%$, $\pm 0.6^\circ$ to 5 kHz; $\pm 5\%$ to 27 kHz. Limit: A1's loop bandwidth
 Range $246\ \Omega$ to $12.1\ \text{M}\Omega$. Limits: TIA compliance below, one LSB of output above; $320\ R_f$ codes in between

The excitation amplitude never enters: Z is the ratio of two records through one converter, so DAC gain and droop cancel.

What sets the EIS error, and what averaging buys

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$$\varepsilon_{|Z|} \approx \underbrace{\frac{R_f/|Z|}{A_{\text{eff}}}}_{\text{systematic}} + \underbrace{\frac{i_n |Z|}{V_1} \sqrt{\frac{2}{NM}}}_{\text{random}}$$

$R_f/|Z|$ feedback resistor over the load: the gain code chosen

A_{eff} effective loop gain, ≈ 76 dB measured (0.66 % at $R_f/R = 40$); $A_{\text{OL}} \approx 83$ dB

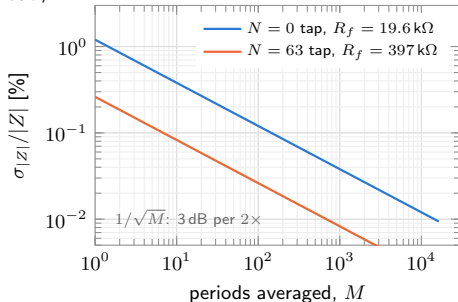
i_n front-end current noise per sample: 273 nA at $N=0$, 61 nA at $N=63$

V_1 excitation fundamental, $\frac{4}{\pi} \frac{A}{2}$

N, M samples per period, periods averaged

The systematic term is fixed for a gain code and load, and the Z_{cal} ratio removes it. The random term is what remains, and it falls as $1/\sqrt{M}$.

Random term against periods averaged, 11 k Ω load, 1 kHz



Calibrating impedance

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$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency}$$

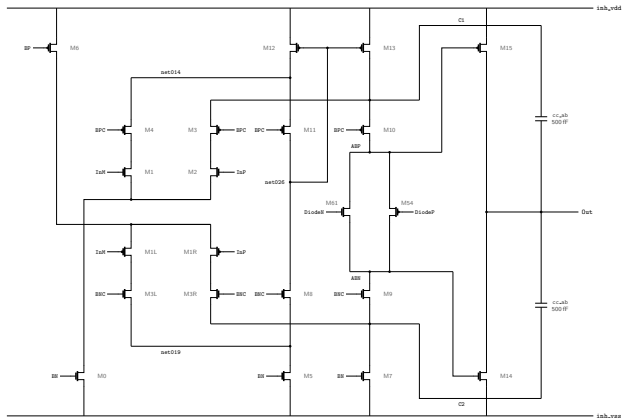
	On-chip poly, per decade	Off-chip standard, at test
Cancels	R_f , its pole, DAC, LSB, loop-gain floor	the same
Known to	$\sim 20\%$ chip to chip	$< 0.1\%$
Run	any time, in the field	once, stored in flash
Good for	trends: fouling is a change in one electrode's Z	absolute ohms

The poly need not be accurate, only stable. One stored number pins the scale; the on-chip ratio carries it through temperature and time.

The class-AB amplifier: one cell, both roles

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complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

The amplifier across corners, with mismatch σ

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Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Open-loop gain	dB	82.77	79.88	83.14	83.37	84.94	1.06
Phase margin	deg	73.52	74.70	74.04	72.86	73.34	1.42
Dominant pole	kHz	1.592	2.286	1.538	1.494	1.225	0.276
Gain-bandwidth product	MHz	22.78	23.19	22.89	22.94	22.59	3.57
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	5.201

Bipolar accuracy through zero

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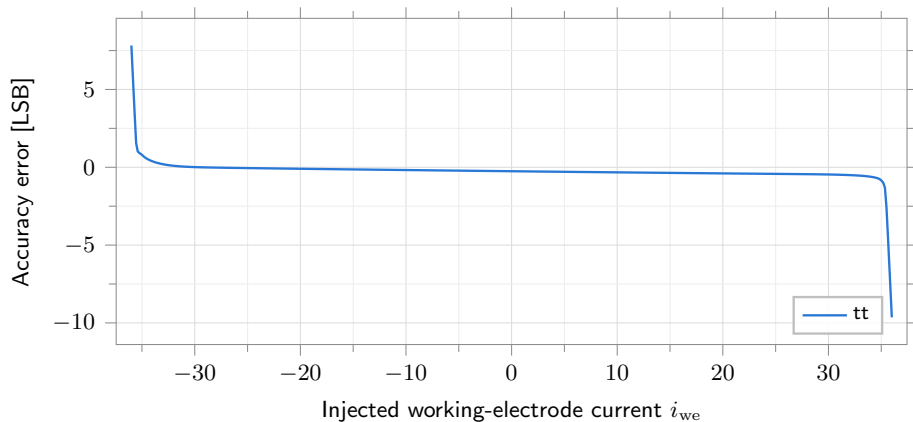
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Programmable feedback resistor: one string, 320 codes

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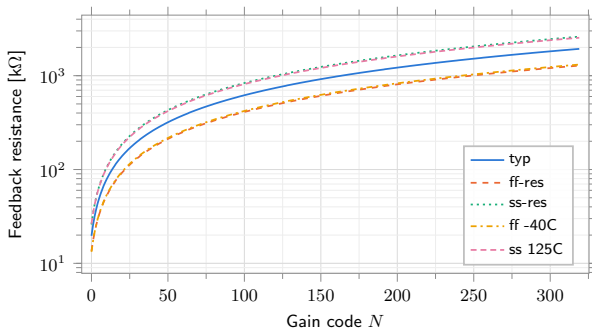
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$$R_f = 18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	±40.7 μA
63	397 kΩ	±2.02 μA
319	1.93 MΩ	±0.41 μA
LSB at 319		0.824 nA

Mismatch σ	σ/typ	
$R(N=0)$	71 Ω	0.36 %
$R(N=319)$	0.7 kΩ	0.04 %
Seam 63→64	504 Ω	6.8 %

10-bit SAR ADC, extracted netlist

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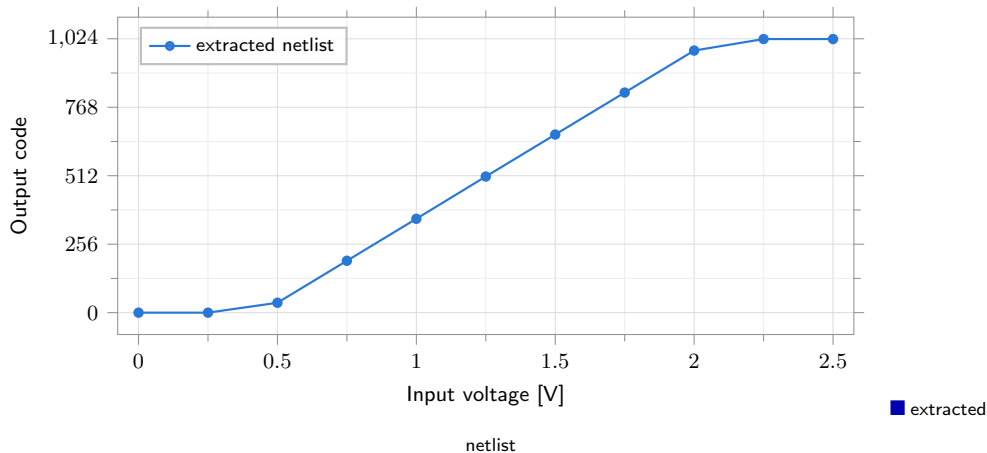
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23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

SAR ADC across corners

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Parameter	Unit	tt	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Analog supply current	μ A	6.36	4.79	5.45	8.44	15.43
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

Reference DAC across corners, with mismatch σ

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Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000001
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.0003
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.610
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.620
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	1.128
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	1.415

One AFE channel, as simulated

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Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

Castalia at a glance

Castalia

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Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Digital, as closed

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Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW

Total	10.72 mW
--------------	-----------------

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Power gating: one switched rail per channel tile

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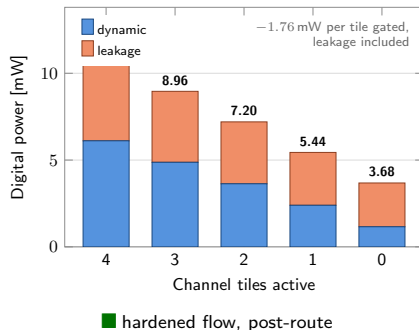
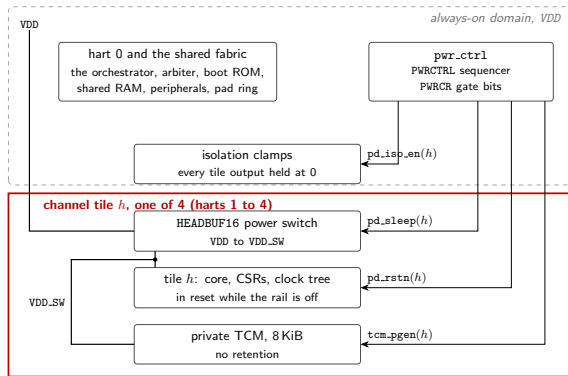
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Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

Vision: the same channel, without the cartridge

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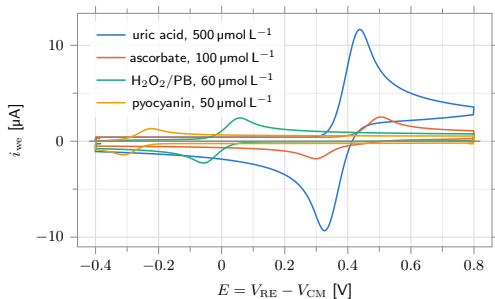
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Already on the die

4 sites	concurrent, each its own DAC, SAR, hart
EIS	electrode state without a second instrument
NFC	harvested boot, battery-free readout
MTCMOS	a parked site costs nothing
NPU	the decision stays on the patient

Left: a small-molecule wound panel, uric acid, ascorbate, H_2O_2 and pyocyanin, through the same channel.

The ovarian panel stays a serum test: CA-125 is an epitope on a 3–5 MDa mucin and does not reach interstitial fluid. The wearable is a different, reagentless panel on the same silicon; the implant is the same plus a fouling problem that EIS can watch but not solve.

Against the field: the ISCAS'26 table, plus Castalia

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Parameter	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Modality	CA/CV/DPV/EIS	CA/CV	CA/CV	CA/CV	CV/EIS
Channels	4, simultaneous	1	1	1, differential	1
Current direction	bipolar	unipolar	bipolar	bipolar	bipolar
Sensitivity (pA)	341 ^a	56.5 ^b	41	0.47	1.7
$I_{\max}$ (μA)	±40.7	+20	±5	±20	-48/+16
Power (mW)	3.68 ^c	3.0 ^d	0.016	9.3	1.0
Process (nm)	65	65	180	350	180
On-chip processor	5 RISC-V harts	1 RISC-V hart	none	none	none
On-chip ML	NPU, 4 modes	NPU, 1 mode	none	none	none

Castalia column is best case and pre-silicon. ^aAFE only: input-referred rms, 0.1–100 Hz, maximum gain, ideal R_f , no electrode; the converter LSB is 0.82 nA. ^bAFE only; ADC quantisation limits the system to 153 pA. ^cDigital only: hart 0 awake, all four channel tiles gated, post-route SS 125 °C 25 MHz; one channel measuring adds 1.76 mW (tile) + 1.14 mW (AFE). ^dMeasured: 2.8 mW digital at 24 MHz + 258 μW AFE.

Still the only entries with a processor and an ML engine on the die; Castalia adds concurrency, both current polarities and EIS.

The fusion runs on the NPU

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$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

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Backup: electrode model validation

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Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Peak current against Randles–Ševčík and ΔE_p against Nicholson [13], reversible couple, $C_{\text{dl}} = 0$ column isolates the faradaic model.

Backup: USPSTF screening grades

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Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.